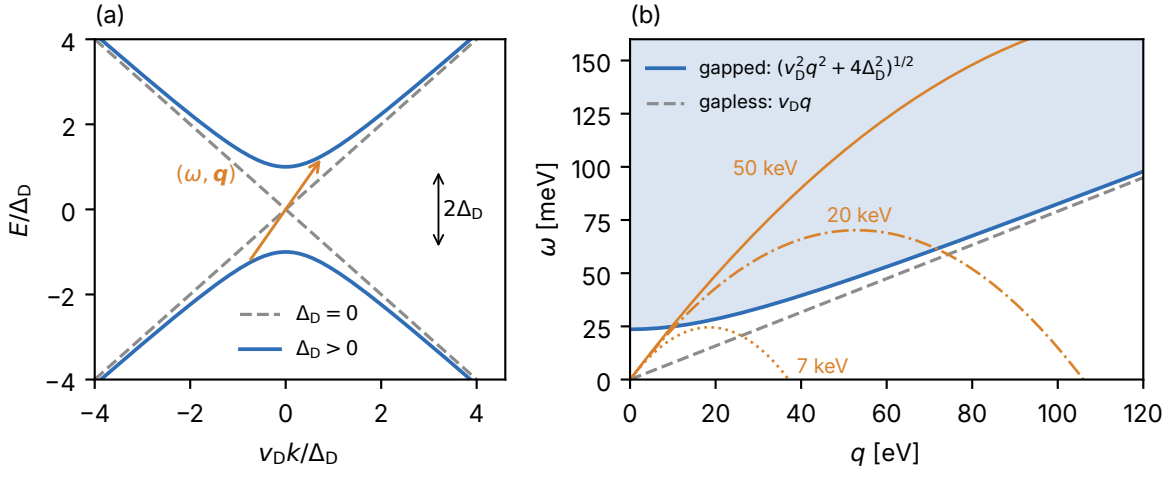
(a)
4
2
0
−2
−4
E/Δ_D
$(\omega, \boldsymbol{q})$
$2\Delta_D$
$\Delta_D = 0$
$\Delta_D > 0$
−4
−2
0
2
4
$v_D k/\Delta_D$
(b)
150
125
100
75
50
25
0
ω [meV]
gapped: $(v_D^2 q^2 + 4\Delta_D^2)^{1/2}$
gapless: $v_D q$
50 keV
20 keV
7 keV
0
20
40
60
80
100
120
q [eV]